# MTH316 2022S1 Exam

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* MTH105 Fundamentals of Mathematics
* MTH107 Calculus I
* MTH108 Calculus II
* MTH316 Multivariable Calculus
* MTH319 Mathematical Methods I
* MTH320 Mathematical Methods II
* MTH207 Linear Algebra
* MTH219 Fundamentals of Statistics and Probability
* MTH220 Statistical Methods and Inference
* MTH355 Basic Mathematical Optimisation

Donaldson Tan

# MTH316

**Timed Online Assignment – January Semester 2022**

# Multivariable Calculus

**Friday, 20 May 2022** **4:00 pm – 6:00 pm**

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**Time allowed: 2 hours**

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**INSTRUCTIONS TO STUDENTS:**

1. This Timed Online Assignment (TOA) contains **SIX (6)** questions and comprises **EIGHT (8)** pages (including cover page).

2. You must answer **ALL** questions.

3. If you have any queries about a question, or believe there is an error in the question, briefly explain your understanding and assumptions about that question before attempting it.

4. You MUST **submit your answers via Canvas (similar to TMA submission) at the end time of this TOA** (as stated on this cover page). The 15 minutes grace period as shown on Canvas is strictly meant for technical issues encountered during submission. Thereafter, you will not be able to submit your answers and you will be considered as having withdrawn from the course. **No appeal will be allowed.**

5. You are to include the following particulars in your submission: TOA Course Code, your Full Name, and SUSS PI Number. Name your submission file as TOACourseCode_FullName_StudentPI. Use underscore and NOT Space. *Example: TOAXYZ_RaphaelLee_T1923161 (omit D/O, S/O)*.

6. Your submission should consist of only one file and **must not exceed 500MB** in size. The file must be a PDF or Microsoft Word file saved in .doc or .docx format. All answers are to be typed. Flowcharts and graphs may be scanned or photographed and embedded in the PDF or Word file provided it does not exceed the file size limit of 500MB.

7. For answers which cannot be typed and require to be hand-written, ensure that question number is clearly stated on each page. All uploaded hand-written answers must be clear, readable and complete. Marks will not be awarded for un-readable or incomplete images.

8. To prevent plagiarism and collusion, your submission will be reviewed by Turnitin. The Turnitin report will only be made available to the marker and you will not be able to view it.

9. The University takes plagiarism and collusion seriously, and your Turnitin report will be examined thoroughly as part of the marking process.

10. Marks will only be awarded if FULL working is shown.

*Answer all questions. (Total 100 Marks)*

**Question 1**

(a) Determine whether

$$\lim_{(x,y)\to(0,0)} \frac{x-\sin(y)}{x+y}$$

exist and evaluate it if it exist.

(6 marks)

(b) Define a function $f : \{(x,y) \in \mathbb{R}^2 \,|\, mx^2 > (m-1)y\} \longrightarrow \mathbb{R}$ by

$$f(x,y) = e^{xy} \ln\left(mx^2 + (1-m)y\right)$$

where $m$ is a positive real constant.

(i) Find $f_x$ and $f_y$ in terms of $m$.

(4 marks)

(ii) Hence, or otherwise, given that $f_x$ and $f_y$ are continuous at the point $(m,m)$ and their values are equal there, determine the possible value(s) of $m$.

(6 marks)

**Question 2**

(a) Consider the two functions of two variables $(x, y)$

$$f(x,y) = 5\tan^{-1}\left(\frac{y}{x}\right) \qquad \text{and} \qquad g(x,y) = \frac{e^{p(x-2)}}{y-p},$$

where $p$ is a real constant and $p \neq 0, 1$. Given that the gradient of $f$ and $g$ are parallel at the point $(2, -1)$, calculate the possible values of $p$.

(6 marks)

(b) To describe a direction, we can make use of the following vector,

$$\mathbf{u} = \cos(\theta)\,\mathbf{i} + \sin(\theta)\,\mathbf{j}, \qquad 0 \leq \theta < 2\pi.$$

Note that $\mathbf{u}$ is a unit vector and $\theta$ describes the direction.

Assume that the function $f$ in question 2(a) describes the height of a mountain at the position $(x, y)$. Peter is climbing the mountain and his current position is at $(5, 5)$.

(i) Find the directions, or equivalently, the range of values of $\theta$ such that if Peter proceeds in those directions, he will be ascending.

(5 marks)

(ii) By writing the directional derivative in the direction of $\mathbf{u}$ in terms of $\theta$, prove that if Peter proceeds in the direction $\frac{\nabla f(5,5)}{|\nabla f(5,5)|}$, he will be ascending at the fastest rate per distance moved.

(5 marks)

**Question 3**

(a) Apply Lagrange Multipliers to find the maximum and minimum of the function

$$f(x,y,z) = e^{xyz}$$

subject to the constraint

$$x^2 + 2y^2 + 3z^2 = 6.$$

(8 marks)

(b) Find the extreme values of the function

$$g(x,y) = 2xy + 2x^2 - 4$$

on the region described by the inequality

$$x^2 + y^2 \leq 4.$$

(10 marks)

## Question 4

Given a 2 dimensional object whose shape is described by a 2 dimensional region $D$, and has a density function $\rho(x,y)$, the center of mass of the object is $(\bar{x},\bar{y})$ where

$$\bar{x} = \frac{\iint_D x\rho(x,y)dA}{\iint_D \rho(x,y)dA} \qquad \text{and} \qquad \bar{y} = \frac{\iint_D y\rho(x,y)dA}{\iint_D \rho(x,y)dA}.$$

Table tennis rackets have different designs to fit the different play styles of players. An "all-round" racket is designed to have higher density towards the center of the racket. Consider region $D$ shown below which is made up of 2 non-overlapping regions $D_1$ and $D_2$ where

$$D_1 = \left\{(x,y)\,\middle|\, x = r\cos(\theta), y = r\sin(\theta), -\frac{\pi}{4} \le \theta \le \frac{5\pi}{4}, 0 \le r \le 6\right\}$$

$$\text{and } D_2 = \left\{(x,y)\,\middle|\, -\frac{6}{\sqrt{2}} \le y \le 0,\ y \le x \le -y\right\}$$

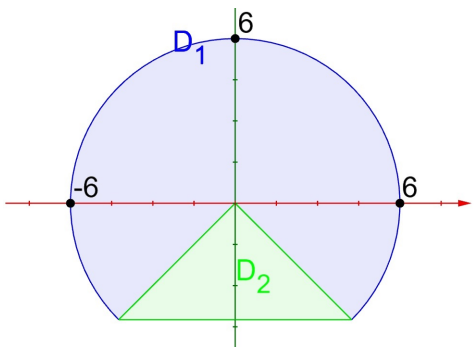



**Figure Q4**

Suppose that region $D$ represents the shape of a table tennis racket and that the density function of the racket is

$$\rho(x,y) = \begin{cases} 49 - x^2 - y^2, & \text{for } (x,y) \in D_1 \\ 49, & \text{for } (x,y) \in D_2. \end{cases}$$

By considering the integration over region $D$ as the sum of integrations over $D_1$ and $D_2$,

(a) evaluate the mass of the racket

$$\iint_D \rho(x,y)dA,$$

(6 marks)

(b) calculate the position of the center of mass of the table tennis racket.

(12 marks)

## Question 5

Let $E$ be the tetrahedron bounded by the planes $x = 0$, $y = 0$, $z = 0$ and $4x + 2y + z = 6$, as shown in Figure Q5 below.

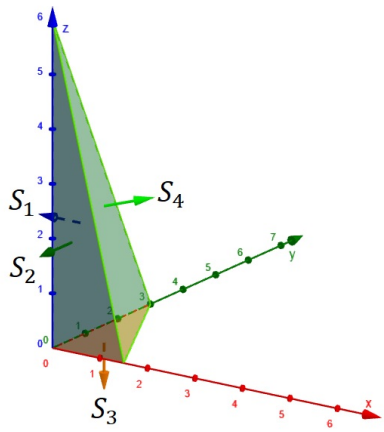



**Figure Q5**

Define a vector field $\mathbf{F}$ as

$$\mathbf{F}(x,y,z) = x^2\,\mathbf{i} + (x+1)y\,\mathbf{j} - 3xz\,\mathbf{k}.$$

(a) Let $S$ be the oriented surface bounding the tetrahedron $E$. By considering the four surfaces that make up $S$,

$$\begin{aligned} S_1 &= S \cap \{(x,y,z) \mid x = 0\} \\ S_2 &= S \cap \{(x,y,z) \mid y = 0\} \\ S_3 &= S \cap \{(x,y,z) \mid z = 0\} \\ S_4 &= S \cap \{(x,y,z) \mid 4x + 2y + z = 6\}, \end{aligned}$$

as shown in Figure Q5 above, compute the surface integral

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$$

**without** using Divergence Theorem where $\mathbf{n}$ is the unit normal vector of $S$ pointing outwards from $E$.

(10 marks)

(b) To verify the theorem, use the Divergence Theorem now to compute the surface integral

$$\iint_S \mathbf{F} \cdot \mathbf{n}\, dS$$

and check that it has the same value as the answer for question 5(a).

(10 marks)

**Question 6**

(a) Show that the vector field

$$\mathbf{F}(x,y,z) = 2xze^{x^2+y^2}\,\mathbf{i} + 2yze^{x^2+y^2}\,\mathbf{j} + e^{x^2+y^2}\,\mathbf{k}$$

is a conservative vector field.

(6 marks)

(b) Hence, or otherwise, compute the line integral

$$\oint_C \mathbf{F}.d\mathbf{r}$$

where $C$ is the closed path

$$\mathbf{r}(t) = 3\cos(t)\,\mathbf{i} + 3\sin(t)\,\mathbf{j} + \sin(t)\,\mathbf{k}, \qquad \text{for } 0 \le t \le 2\pi.$$

(6 marks)

—— END OF PAPER ——

1(a) let $f(x,y) = \dfrac{x - \sin(y)}{x+y}$

moving along x-axis

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{x\to 0} f(x,0)$$

$$= \lim_{x\to 0} \frac{x - \sin(0)}{x+0}$$

$$= 1$$

moving along y-axis

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{y\to 0} f(0,y)$$

$$= \lim_{y\to 0} \frac{-\sin y}{y}$$

$\frac{0}{0}$ form, apply L'Hopital Rule

$$= \lim_{y\to 0} \frac{-\cos y}{1}$$

$$= -\cos(0)$$

$$= -1$$

The limit doesn't exist

(b) $f(x,y) = e^{xy} \ln[mx^2 + (1-m)y]$

$$f_x = ye^{xy} \ln[mx^2 + (1-m)y] + e^{xy} \cdot \frac{2mx}{mx^2 + (1-m)y}$$

$$f_y = xe^{xy} \ln[mx^2 + (1-m)y] + e^{xy} \cdot \frac{1-m}{mx^2 + (1-m)y}$$

$$f_x(m,m) = f_y(m,m)$$

$$me^{m^2} \ln[m^3 - m^2 + m] + e^{m^2} \cdot \frac{2m^2}{m^3 - m^2 + m} = me^{m^2} \ln[m^3 - m^2 + m] + e^{m^2} \cdot \frac{1-m}{m^3 - m^2 + m}$$

$$\frac{2m^2 e^{m^2}}{m^3 - m^2 + m} = \frac{(1-m)e^{m^2}}{m^3 - m^2 + m}$$

$$\frac{2m^2 e^{m^2} - (1-m)e^{m^2}}{m^3 - m^2 + m} = 0$$

$$\frac{(2m^2 + m - 1)e^{m^2}}{m^3 - m^2 + m} = 0$$

$$\frac{(2m-1)(m+1)e^{m^2}}{m^3 - m^2 + m} = 0$$

$$(2m-1)(m+1)e^{m^2} = 0$$

$\therefore m = \frac{1}{2}$ or $-1$ (reject)

2(a) $f(x,y) = 5\tan^{-1}\left(\frac{y}{x}\right)$

$$\frac{\partial f}{\partial x} = 5 \cdot \frac{-\frac{y}{x^2}}{1+\left(\frac{y}{x}\right)^2} = -\frac{5y}{x^2+y^2}$$

$$\frac{\partial f}{\partial y} = 5 \cdot \frac{\frac{1}{x}}{1+\left(\frac{y}{x}\right)^2} = \frac{5x}{x^2+y^2}$$

$$\nabla f(2,-1) = \begin{bmatrix} -\frac{5(-1)}{2^2+(-1)^2} \\ \frac{5(2)}{2^2+(-1)^2} \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$g(x,y) = \frac{e^{p(x-2)}}{y-p}$$

$$\frac{\partial g}{\partial x} = \frac{pe^{p(x-2)}}{y-p}$$

$$\frac{\partial g}{\partial y} = -\frac{e^{p(x-2)}}{(y-p)^2}$$

$$\nabla g(2,-1) = \begin{bmatrix} \frac{pe^0}{-(-p)} \\ -\frac{e^0}{(-(-p))^2} \end{bmatrix} = \begin{bmatrix} -\frac{p}{p+1} \\ -\frac{1}{(p+1)^2} \end{bmatrix}$$

since $\nabla f(2,-1) \nparallel \nabla g(2,-1)$

$$\frac{-\frac{p}{p+1}}{-\frac{1}{(p+1)^2}} = \frac{1}{2}$$

$$\frac{2p}{p+1} = \frac{1}{(p+1)^2}$$

$$2p(p+1)^2 = p+1$$

$$2p(p+1)^2 - (p+1) = 0$$

$$(p+1)[2p(p+1)-1] = 0$$

$$(p+1)(2p^2+2p-1) = 0$$

case $2p^2+2p-1=0$ :

$$p^2+p-\frac{1}{2} = 0$$

$$\left(p+\frac{1}{2}\right)^2 - \frac{3}{4} = 0$$

$$\left(p+\frac{1}{2}\right)^2 = \frac{3}{4}$$

$$\therefore p = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}.$$

case $p+1=0$ :

$p=-1$ (reject)

[invalid denominator]

2(b) i. $u = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$ where $0 \le \theta \le 2\pi$

$$\nabla f(5,5) = \begin{bmatrix} -\frac{5(5)}{5^2+5^2} \\ \frac{5(5)}{5^2+5^2} \end{bmatrix} = \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

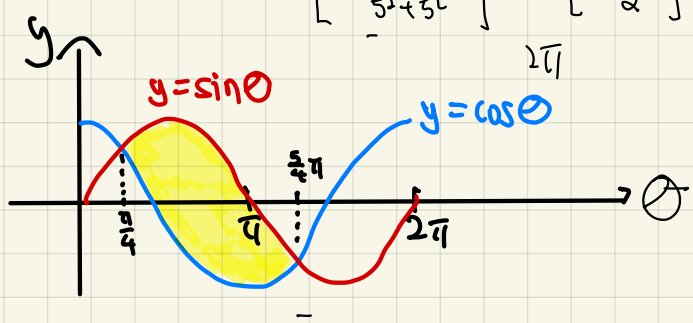
let $D_u f(5,5) > 0$

$$\nabla f(5,5) \cdot \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} > 0$$

$$-\frac{1}{2}\cos\theta + \frac{1}{2}\sin\theta > 0$$

$$\sin\theta - \cos\theta > 0$$

$$\sin\theta > \cos\theta$$




$$\therefore \frac{\pi}{4} < \theta < \frac{5}{4}\pi$$

ii. $D_u f(5,5) = \nabla f(5,5) \cdot \underset{\sim}{u}$

$$= \begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} \cdot \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$$= -\frac{1}{2}\cos\theta + \frac{1}{2}\sin\theta$$

$$= \sqrt{\left(-\frac{1}{2}\right)^2 + \left(\frac{1}{2}\right)^2}\, \sin\left(\theta + \tan^{-1}\frac{-\frac{1}{2}}{\frac{1}{2}}\right)$$

$$= \frac{1}{\sqrt{2}}\sin(\theta + \tan^{-1} -1)$$

$$= |\nabla f(5,5)| \sin\left(\theta - \frac{\pi}{4}\right)$$

when $D_u f(5,5)$ is maximum

$$|\nabla f(5,5)| \sin\left(\theta - \frac{\pi}{4}\right) = |\nabla f(5,5)|$$

$$\sin\left(\theta - \frac{\pi}{4}\right) = 1$$

$$\theta = \frac{\pi}{4} + \sin^{-1} 1 = \frac{\pi}{4} + \frac{\pi}{2} = \frac{3}{4}\pi$$

$u$ for max $D_f u(5,5)$ $= \begin{bmatrix} \cos\frac{3}{4}\pi \\ \sin\frac{3}{4}\pi \end{bmatrix}$

$$= \begin{bmatrix} -\frac{\sqrt{2}}{2} \\ \frac{\sqrt{2}}{2} \end{bmatrix}$$

$$= \sqrt{2}\begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$= \frac{1}{\frac{1}{\sqrt{2}}}\begin{bmatrix} -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix} = \frac{\nabla f(5,5)}{|\nabla f(5,5)|}$$

3(a) optimise $f(x,y,z) = e^{xyz}$

subject to $g(x,y,z) = 6$ where $g(x,y,z) = x^2 + 2y^2 + 3z^2$

$\frac{\partial f}{\partial x} = yz\,e^{xyz}$ $\quad \frac{\partial f}{\partial y} = xz\,e^{xyz}$ $\quad \frac{\partial f}{\partial z} = xy\,e^{xyz}$

$\frac{\partial g}{\partial x} = 2x$ $\quad \frac{\partial g}{\partial y} = 4y$ $\quad \frac{\partial g}{\partial z} = 6z$

let $\nabla f = \lambda \nabla g$ where $\lambda$ is some constant

$$\Rightarrow \lambda = \frac{yze^{xyz}}{2x} = \frac{xze^{xyz}}{4y} = \frac{xye^{xyz}}{6z}$$

$$\frac{yze^{xyz}}{2x} - \frac{xze^{xyz}}{4y} = 0 \Rightarrow \frac{ze^{xyz}}{2}\left[\frac{y}{x} - \frac{x}{2y}\right] = 0$$

$$\frac{xze^{xyz}}{4y} - \frac{xye^{xyz}}{6z} = 0 \Rightarrow \frac{xe^{xyz}}{2}\left[\frac{z}{2y} - \frac{y}{3z}\right] = 0$$

$$\frac{yze^{xyz}}{2x} - \frac{xye^{xyz}}{6z} = 0 \Rightarrow \frac{ye^{xyz}}{2}\left[\frac{z}{x} - \frac{x}{3z}\right] = 0$$

$$\frac{y}{x} - \frac{x}{2y} = 0 \Rightarrow \frac{y}{x} = \frac{x}{2y} \Rightarrow x^2 = 2y^2$$

$$\frac{z}{2y} - \frac{y}{3z} = 0 \Rightarrow \frac{z}{2y} = \frac{y}{3z} \Rightarrow 2y^2 = 3z^2$$

$$\frac{z}{x} - \frac{x}{3z} = 0 \Rightarrow \frac{z}{x} = \frac{x}{3z} \Rightarrow x^2 = 3z^2$$

$$x^2 = 2y^2 = 3z^2$$

$$x^2 + 2y^2 + 3z^2 = 6$$

$$3x^2 = 6$$

$$x^2 = 2$$

$$y^2 = \frac{x^2}{2} = 1$$

$$z^2 = \frac{x^2}{3} = \frac{2}{3}$$

$$(xyz)^2 = 2(1)\left(\frac{2}{3}\right) = \frac{4}{3}$$

$$xyz = \pm\sqrt{\frac{4}{3}}$$

$\therefore$ max $f(x,y,z) = e^{\sqrt{\frac{4}{3}}}$

$\therefore$ min $f(x,y,z) = e^{-\sqrt{\frac{4}{3}}}$

3(b) $g(x,y) = 2xy + 2x^2 - 4$

$\frac{\partial g}{\partial x} = 2y + 4x \qquad \frac{\partial g}{\partial y} = 2x \qquad \frac{\partial^2 g}{\partial x \partial y} = 2$

$\frac{\partial^2 g}{\partial x^2} = 4 \qquad \frac{\partial^2 g}{\partial y^2} = 0$

let $\frac{\partial g}{\partial x} = \frac{\partial g}{\partial y} = 0$

$\frac{\partial g}{\partial y} = 0 \Rightarrow x = 0$

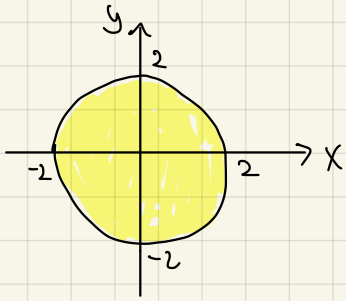
$\frac{\partial g}{\partial x} = 0 \Rightarrow y = -2x = 0$

$(0,0)$ is a critical point

$\left(\frac{\partial^2 g}{\partial x}\right)\left(\frac{\partial^2 g}{\partial y}\right) - \left(\frac{\partial^2 g}{\partial x \partial y}\right)^2 = 4(0) - 2^2 = -4 < 0$

$\therefore (0,0)$ is a saddle point

$g(0,0) = -4$




search the circumference for extreme value

$\left.\begin{aligned} x(\theta) &= 2\cos\theta \\ y(\theta) &= 2\sin\theta \end{aligned}\right\} \quad 0 \le \theta \le 2\pi$

$g(\theta) = g(2\cos\theta, 2\sin\theta)$

$= 8\sin\theta\cos\theta + 8\cos^2\theta - 4$

$= 4\sin 2\theta + 4\cos 2\theta$

$g'(\theta) = 8\cos 2\theta - 8\sin 2\theta$

let $g'(\theta) = 0$

$8\sin 2\theta = 8\cos 2\theta$

$\tan 2\theta = 1$

$2\theta = \frac{\pi}{4} + n\pi$ where $n = 0,1,2,3$ $\quad [0 \le 2\theta \le 4\pi]$

$\theta = \frac{\pi}{8} + \frac{n}{2}\pi$

$= \left(\frac{1}{8} + \frac{n}{2}\right)\pi$

$= \frac{\pi}{8}, \frac{5}{8}\pi, \frac{9}{8}\pi, \frac{13}{8}\pi$

$g\left(\frac{\pi}{8}\right) = g\left(\frac{9}{8}\pi\right) = 4\sqrt{2}$

$g\left(\frac{5}{8}\pi\right) = g\left(\frac{13}{8}\pi\right) = -4\sqrt{2}$

$\therefore \min g(x,y) = -4\sqrt{2}$

$\therefore \max g(x,y) = 4\sqrt{2}$

4(a) $\iint_{D_1} \rho(x,y)\, dA = \int_{-\frac{\pi}{4}}^{\frac{5}{4}\pi} \int_0^6 (49-r^2) r \, dr\, d\theta$

$= \int_{-\frac{\pi}{4}}^{\frac{5}{4}\pi} \int_0^6 49r - r^3 \, dr\, d\theta$

$= \int_{-\frac{\pi}{4}}^{\frac{5}{4}\pi} \left[ \frac{49}{2} r^2 - \frac{1}{4} r^4 \right]_0^6 d\theta$

$= \int_{-\frac{\pi}{4}}^{\frac{5}{4}\pi} 558 \, d\theta$

$= \left[ 558\theta \right]_{-\frac{\pi}{4}}^{\frac{5}{4}\pi}$

$= 837\pi$

$\iint_{D_2} \rho(x,y)\, dA = \int_{-\frac{6}{\sqrt{2}}}^{0} \int_y^{-y} 49 \, dx\, dy$

$= \int_{-\frac{6}{\sqrt{2}}}^{0} \left[ 49x \right]_y^{-y} dy$

$= \int_{-\frac{6}{\sqrt{2}}}^{0} -98y \, dy$

$= \left[ -49y^2 \right]_{-\frac{6}{\sqrt{2}}}^{0}$

$= 882$

$\therefore$ mass $m = \iint_{D_1} \rho(x,y)\, dA + \iint_{D_2} \rho(x,y)\, dA$

$= 837\pi + 882$

(b) By symmetry, $\bar{x} = 0$

$\iint_{D_1} y\rho(x,y)\, dA = \int_{-\frac{\pi}{4}}^{\frac{5}{4}\pi} \int_0^6 (49-r^2)(r\sin\theta)\, r \, dr\, d\theta$

$= \int_{-\frac{\pi}{4}}^{\frac{5}{4}\pi} \sin\theta \int_0^6 49r^2 - r^4 \, dr\, d\theta$

$= \int_{-\frac{\pi}{4}}^{\frac{5}{4}\pi} \sin\theta \left[ \frac{49}{3} r^3 - \frac{1}{5} r^5 \right]_0^6 d\theta$

$= \int_{-\frac{\pi}{4}}^{\frac{5}{4}\pi} \frac{9864}{5} \sin\theta \, d\theta$

$= \frac{9864}{5} \left[ -\cos\theta \right]_{-\frac{\pi}{4}}^{\frac{5}{4}\pi}$

$= \frac{9864\sqrt{2}}{5}$

$\iint_{D_2} y\rho(x,y)\, dA = \int_{-\frac{6}{\sqrt{2}}}^{0} \int_y^{-y} 49y \, dx\, dy$

$= \int_{-\frac{6}{\sqrt{2}}}^{0} 49y \left[ x \right]_y^{-y} dy$

$= \int_{-\frac{6}{\sqrt{2}}}^{0} -98y^2 \, dy$

$= \left[ -\frac{98}{3} y^3 \right]_{-\frac{6}{\sqrt{2}}}^{0}$

$= -1764\sqrt{2}$

$\iint_D y\rho(x,y)\, dA = \iint_{D_1} y\rho(x,y)\, dA + \iint_{D_2} y\rho(x,y)\, dA$

$= \frac{9864\sqrt{2}}{5} - 1764\sqrt{2}$

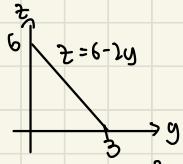
$= \frac{1044}{5}\sqrt{2}$

$\bar{y} = \frac{1}{m} \iint_D y\, \rho(x,y)\, dA$

$= \frac{1044\sqrt{2}}{5(882+837\pi)}$

$\therefore$ centre of mass $= (\bar{x}, \bar{y}) = \left(0, \frac{1044\sqrt{2}}{5(882+837\pi)}\right)$

5(a) consider surface $S_1$




$$\underset{\sim}{n} = \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix}$$

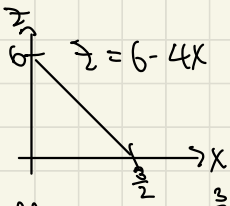
$$\iint_{S_1} F \cdot dS = \int_0^3 \int_0^{6-2y} F(0,y,z) \cdot \underset{\sim}{n} \, dx\,dy$$

$$= \int_0^3 \int_0^{6-2y} \begin{bmatrix} 0 \\ y \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -1 \\ 0 \\ 0 \end{bmatrix} dx\,dy$$

$$= \int_0^3 \int_0^{6-2y} 0 \, dx\,dy$$

$$= 0$$

consider surface $S_2$




$$\underset{\sim}{n} = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix}$$

$$\iint_{S_2} F \cdot dS = \int_0^{\frac{3}{2}} \int_0^{6-3x} F(x,0,z) \cdot \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} dy\,dx$$

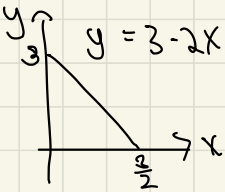
$$= \int_0^2 \int_0^{6-3x} \begin{bmatrix} x^2 \\ 0 \\ -3xz \end{bmatrix} \cdot \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix} dy\,dx$$

$$= \int_0^2 \int_0^{6-3x} 0 \, dy\,dx$$

$$= 0$$

consider surface $S_3$

$$\underset{\sim}{n} = \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix}$$




$$\iint_{S_3} F \cdot dS = \int_0^2 \int_0^{3-2x} F(x,y,0) \cdot \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} dy\,dx$$

$$= \int_0^2 \int_0^{3-2x} \begin{bmatrix} x^2 \\ (x+1)y \\ 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ -1 \end{bmatrix} dy\,dx$$

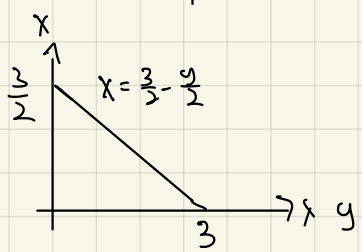
$$= \int_0^2 \int_0^{3-2x} 0 \, dy\,dx = 0$$

5(a) consider surface $S_4$

$$r(x,y) = \begin{bmatrix} x \\ y \\ 6-4x-2y \end{bmatrix}$$

$$r_x(x,y) = \begin{bmatrix} 1 \\ 0 \\ -4 \end{bmatrix} \qquad r_y = \begin{bmatrix} 0 \\ 1 \\ -2 \end{bmatrix}$$

$$r_x \times r_y = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ 1 & 0 & -4 \\ 0 & 1 & -2 \end{vmatrix} = \begin{bmatrix} 0-(-4) \\ -(-2-0) \\ 1-0 \end{bmatrix} = \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix}$$




$$\iint_{S_4} F \cdot \underset{\sim}{n}\, dS = \int_0^3 \int_0^{\frac{3}{2}-\frac{y}{2}} F(x,y,6-4x-2y) \cdot \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} dx\, dy$$

$$= \int_0^3 \int_0^{\frac{3}{2}-\frac{y}{2}} \begin{bmatrix} x^2 \\ (x+1)y \\ -3x(1-4x-2y) \end{bmatrix} \cdot \begin{bmatrix} 4 \\ 2 \\ 1 \end{bmatrix} dx\, dy$$

$$= \int_0^3 \int_0^{\frac{3}{2}-\frac{y}{2}} 16x^2 + 8xy - 18x + 2y \quad dx\, dy$$

$$= \int_0^3 \left[ \frac{16}{3}x^3 + 4x^2y - 9x^2 + 2xy \right]_{x=0}^{x=\frac{3}{2}-\frac{y}{2}}$$

$$= \int_0^3 \frac{y^3}{3} - \frac{13}{4}y^2 + \frac{15}{2}y - \frac{9}{4}\, dy$$

$$= \left[ \frac{y^4}{12} - \frac{13}{12}y^3 + \frac{15}{4}y^2 - \frac{9}{4}y \right]_0^3$$

$$= \frac{9}{2}$$

$$\therefore \iint_S F \cdot \underset{\sim}{n}\, dS = \iint_{S_1} F \cdot \underset{\sim}{n}\, dS + \iint_{S_2} F \cdot \underset{\sim}{n}\, dS + \iint_{S_3} F \cdot \underset{\sim}{n}\, dS + \iint_{S_4} F \cdot \underset{\sim}{n}\, dS$$

$$= 0 + 0 + 0 + \frac{9}{2}$$

$$= \frac{9}{2}$$

5(b)

$$\text{div } F = \nabla \cdot F$$

$$= \begin{bmatrix} \frac{\partial}{\partial x} \\ \frac{\partial}{\partial y} \\ \frac{\partial}{\partial z} \end{bmatrix} \cdot \begin{bmatrix} x^2 \\ (x+1)y \\ -3xz \end{bmatrix}$$

$$= 2x + (x+1) - 3x$$

$$= 1$$

$$\iint_S F \cdot n \, dS = \iiint_E \text{div } F \, dV$$

$$= \int_0^{\frac{3}{2}} \int_0^{3-2x} \int_0^{6-4x-2y} 1 \, dz \, dy \, dx$$

$$= \int_0^{\frac{3}{2}} \int_0^{3-2x} 6-4x-2y \, dy \, dx$$

$$= \int_0^{\frac{3}{2}} \left[ 6y - 4xy - y^2 \right]_{y=0}^{y=3-2x} dx$$

$$= \int_0^{\frac{3}{2}} -(3-2x)^2 + (3-2x)(6-4x) \, dx$$

$$= \left[ \frac{4}{3}x^3 - 6x^2 + 9x \right]_0^{\frac{3}{2}}$$

$$= \frac{9}{2}$$

6. $F(x,y,z) = \begin{bmatrix} 2xze^{x^2+y^2} \\ 2yze^{x^2+y^2} \\ e^{x^2+y^2} \end{bmatrix}$

(a)

$\Rightarrow F(x,y,z)$ is continuous in $\mathbb{R}^3$

$$\text{curl } F = \begin{vmatrix} \underset{\sim}{i} & \underset{\sim}{j} & \underset{\sim}{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xze^{x^2+y^2} & 2yze^{x^2+y^2} & e^{x^2+y^2} \end{vmatrix}$$

$$= \begin{bmatrix} 2ye^{x^2+y^2} - 2ye^{x^2+y^2} \\ -(2xe^{x^2+y^2} - 2xe^{x^2+y^2}) \\ 4xyze^{x^2+y^2} - 4xyze^{x^2+y^2} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$\therefore F(x,y,z)$ is conservative

(b) closed path in a conservative field

$\therefore \oint_C F \cdot dr = 0$